

GED Math Practice Test 1

PART A — QUESTIONS

GED-MATH1-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A grocery store sells apples for \$0.80 each. What is the cost of 5 apples?

A) \$3.20

B) \$3.60

C) \$4.00

D) \$4.80

GED-MATH1-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of $3x - 7$ when $x = 5$?

A) 8

B) 10

C) 12

D) 22

GED-MATH1-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A recipe uses $\frac{3}{4}$ cup of oats. If Maya makes half of the recipe, how many cups of oats will she use?

A) $\frac{1}{4}$ cup

B) $\frac{3}{8}$ cup

C) $\frac{1}{2}$ cup

D) $\frac{2}{3}$ cup

GED-MATH1-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to $4(y + 3) - 2y$?

A) $2y + 3$

B) $2y + 12$

- C) $4y + 1$
D) $6y + 12$

GED-MATH1-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A jacket originally costs \$60. It is discounted by 25%. What is the sale price?

- A) \$15
B) \$35
C) \$45
D) \$75

GED-MATH1-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A taxi company charges a flat fee of \$4 plus \$2.25 per mile. Which expression represents the cost, in dollars, for a ride of m miles?

- A) $4m + 2.25$
B) $2.25m + 4$
C) $6.25m$
D) $4(m + 2.25)$

GED-MATH1-007

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular garden is 18 feet long and 11 feet wide. What is the area of the garden in square feet? Type your answer as an integer.

GED-MATH1-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $C = 35 + 12h$ gives the total cost C , in dollars, to rent a kayak for h hours. In this formula, 35 represents [D1], and 12 represents [D2].

Dropdowns:

- D1) the cost per hour | the fixed starting cost | the total number of hours
D2) the cost per hour | the fixed starting cost | the total rental cost

GED-MATH1-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A car travels 156 miles using 6 gallons of gas. What is the car's fuel efficiency in miles per gallon?

A) 24

B) 25

C) 26

D) 28

GED-MATH1-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for n: $5n + 8 = 43$.

A) 5

B) 7

C) 9

D) 11

GED-MATH1-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A school club has 18 boys and 24 girls. What is the ratio of boys to girls in simplest form?

A) 3:4

B) 4:3

C) 6:8

D) 18:42

GED-MATH1-012

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which equations have $x = 6$ as a solution?

A) $2x + 5 = 17$

B) $3x - 4 = 14$

C) $x/2 + 8 = 10$

D) $4x + 1 = 21$

E) $5x - 6 = 30$

GED-MATH1-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows the number of books sold by a bookstore during four days.

Day: Monday, Tuesday, Wednesday, Thursday

Books sold: 42, 38, 55, 45

What is the mean number of books sold per day?

A) 42

B) 44

C) 45

D) 48

GED-MATH1-014

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A line has equation $y = 3x - 8$. What is the value of y when $x = 12$?

Type your answer as an integer.

GED-MATH1-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A map uses a scale of 1 inch = 12 miles. Two towns are 3.5 inches apart on the map. What is the actual distance between the towns?

A) 15.5 miles

B) 36 miles

C) 42 miles

D) 48 miles

GED-MATH1-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A phone plan charges \$30 per month plus \$0.10 per text message. Which equation can be used to find t , the number of text messages, if the monthly bill is \$47?

A) $30t + 0.10 = 47$

B) $30 + 0.10t = 47$

C) $0.10(t + 30) = 47$

D) $47 + 0.10t = 30$

GED-MATH1-017

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A storage container is a rectangular prism with length 6 feet, width 4 feet, and height 3 feet. Which statements are true?

- A) The volume is 72 cubic feet.
- B) The volume is 24 cubic feet.
- C) The area of the base is 24 square feet.
- D) The area of the base is 13 square feet.
- E) The height equals the area of the base divided by the volume.

GED-MATH1-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which inequality represents “twice a number is at least 18”?

- A) $2x > 18$
- B) $2x \geq 18$
- C) $x + 2 \leq 18$
- D) $x/2 \geq 18$

GED-MATH1-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A nurse records the following resting heart rates, in beats per minute:

68, 72, 75, 70, 80

What is the median heart rate?

- A) 70
- B) 72
- C) 73
- D) 75

GED-MATH1-020

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for x: $4x - 9 = 31$.

Type your answer as an integer.

GED-MATH1-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular parking lot has a perimeter of 190 meters. Its length is 60 meters. What is its width?

A) 25 meters

B) 35 meters

C) 70 meters

D) 130 meters

GED-MATH1-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The value of a computer decreases by 15% each year. If the computer is worth \$900 today, which expression gives its value after 2 years?

A) $900(0.15)^2$

B) $900(0.85)^2$

C) $900(1.15)^2$

D) $900 - 0.85^2$

GED-MATH1-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A bag contains 5 red marbles, 3 blue marbles, and 2 green marbles. If one marble is chosen at random, what is the probability that it is blue?

A) $\frac{1}{5}$

B) $\frac{3}{10}$

C) $\frac{1}{2}$

D) $\frac{3}{7}$

GED-MATH1-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A linear function is shown in the table.

x: 1, 2, 3, 4

y: 7, 11, 15, 19

Which equation represents the function?

A) $y = 3x + 4$

- B) $y = 4x + 3$
- C) $y = 5x + 2$
- D) $y = 7x$

GED-MATH1-025

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A restaurant bill is \$48 before tax. The sales tax rate is 8%. What is the total bill, including tax?
Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH1-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve the system of equations.

$$x + y = 19$$

$$x - y = 5$$

What is the value of x ?

- A) 7
- B) 10
- C) 12
- D) 14

GED-MATH1-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A ladder leans against a wall. The bottom of the ladder is 5 feet from the wall, and the top reaches a point 12 feet up the wall. What is the length of the ladder?

- A) 13 feet
- B) 15 feet
- C) 17 feet
- D) 60 feet

GED-MATH1-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A function is defined by $f(x) = 2x^2 + 1$. The value of $f(3)$ is [D1].

Dropdowns:

D1) 13 | 18 | 19

GED-MATH1-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A customer buys 2.5 pounds of trail mix at \$6.40 per pound. How much does the customer pay?

- A) \$8.90
- B) \$12.80
- C) \$16.00
- D) \$18.90

GED-MATH1-030

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which statements are true about the line $y = -2x + 9$?

- A) The slope is -2 .
- B) The y-intercept is 9.
- C) The line rises from left to right.
- D) The x-intercept is 9.
- E) The slope is 9.

GED-MATH1-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A warehouse ships 840 boxes in 7 equal truckloads. How many boxes are in each truckload?

- A) 105
- B) 110
- C) 120
- D) 147

GED-MATH1-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for y: $3(y - 4) = 2y + 9$.

- A) 9
- B) 13

- C) 17
- D) 21

GED-MATH1-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cylindrical can has a radius of 4 centimeters and a height of 10 centimeters. Using $\pi \approx 3.14$, what is the volume of the can?

- A) 125.6 cm³
- B) 251.2 cm³
- C) 502.4 cm³
- D) 1,004.8 cm³

GED-MATH1-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve the literal equation $A = 1/2bh$ for h .

- A) $h = A/(2b)$
- B) $h = 2A/b$
- C) $h = b/(2A)$
- D) $h = 2b/A$

GED-MATH1-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A membership website charges \$18 to join and \$7 per month. A second website charges no joining fee and \$10 per month. After how many months will the total cost be the same?

- A) 3 months
- B) 4 months
- C) 6 months
- D) 9 months

GED-MATH1-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A store manager orders 36 boxes of paper. Each box contains 500 sheets. The manager uses 7,250 sheets during the first month. How many sheets are left?

- A) 9,750
- B) 10,250

- C) 10,750
- D) 18,000

GED-MATH1-037

Section: Math

Type: drag_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A) $y = 5x + 2$
- B) $y = 2x + 5$
- C) $y = 5x - 2$
- D) $y = -5x + 2$

Targets:

- T1) A line with slope 5 and y-intercept 2
- T2) A line with slope 2 and y-intercept 5
- T3) A line with slope -5 and y-intercept 2

GED-MATH1-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

An apartment uses 680 kilowatt-hours of electricity in April and 748 kilowatt-hours in May. What is the percent increase from April to May?

- A) 6%
- B) 8%
- C) 10%
- D) 12%

GED-MATH1-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which equation has the solution $x = -3$?

- A) $x + 8 = 12$
- B) $2x - 5 = -11$
- C) $4x + 1 = 9$
- D) $x/3 = 3$

GED-MATH1-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A ball is dropped from a height of 80 inches. After each bounce, it reaches 60% of its previous height. What height will it reach after the second bounce?

- A) 28.8 inches
- B) 36 inches
- C) 48 inches
- D) 64 inches

GED-MATH1-041

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A city survey recorded the number of people in 6 households:

2, 3, 4, 4, 5, 6

Which statements are true?

- A) The range is 4.
- B) The median is 4.
- C) The mean is 5.
- D) The range is 6.
- E) The median is 4.5.

GED-MATH1-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function $g(x) = -4x + 15$, what is $g(-2)$?

- A) 7
- B) 13
- C) 17
- D) 23

GED-MATH1-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A small business makes notebooks. The total cost C , in dollars, to make n notebooks is $C = 120 + 2.75n$. What does the number 120 represent?

- A) The cost to make each notebook
- B) The fixed cost before making any notebooks
- C) The number of notebooks made
- D) The total cost after making 120 notebooks

GED-MATH1-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which table represents a linear relationship with a constant rate of change of 6?

A) x: 0, 1, 2, 3

y: 4, 10, 16, 22

B) x: 0, 1, 2, 3

y: 6, 10, 14, 18

C) x: 0, 1, 2, 3

y: 2, 6, 12, 20

D) x: 0, 1, 2, 3

y: 8, 14, 19, 23

GED-MATH1-045

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation $p = 0.08s$ gives the sales tax p on an item with price s . Which equation solves for s ?

A) $s = p + 0.08$

B) $s = p - 0.08$

C) $s = p/0.08$

D) $s = 0.08/p$

GED-MATH1-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A community center has a rectangular room that is 32 feet long and 24 feet wide. Carpet costs \$2.75 per square foot. What is the total cost to carpet the room?

A) \$1,344

B) \$1,536

C) \$2,112

D) \$2,400

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH1-001 — Correct: C — Correct Answer: \$4.00 — Explanation: Multiply \$0.80 by 5 to get \$4.00.

GED-MATH1-002 — Correct: A — Correct Answer: 8 — Explanation: Substitute $x = 5$: $3(5) - 7 = 15 - 7 = 8$.

GED-MATH1-003 — Correct: B — Correct Answer: $3/8$ cup — Explanation: Half of $3/4$ is $3/4 \div 2 = 3/8$.

GED-MATH1-004 — Correct: B — Correct Answer: $2y + 12$ — Explanation: Distribute first: $4(y + 3) - 2y = 4y + 12 - 2y = 2y + 12$.

GED-MATH1-005 — Correct: C — Correct Answer: \$45 — Explanation: A 25% discount on \$60 is \$15, so the sale price is $\$60 - \$15 = \$45$.

GED-MATH1-006 — Correct: B — Correct Answer: $2.25m + 4$ — Explanation: The total cost is the \$4 flat fee plus \$2.25 for each mile, or $2.25m + 4$.

GED-MATH1-007 — Correct: 198 — Tolerance: exact — Correct Answer: 198 — Explanation: Area = length $\times$ width = $18 \times 11 = 198$ square feet.

GED-MATH1-008 — Correct: D1=the fixed starting cost; D2=the cost per hour — Correct Answer: D1=the fixed starting cost; D2=the cost per hour — Explanation: The constant 35 is the starting cost, and the coefficient 12 is the hourly cost.

GED-MATH1-009 — Correct: C — Correct Answer: 26 — Explanation: Divide miles by gallons: $156 \div 6 = 26$ miles per gallon.

GED-MATH1-010 — Correct: B — Correct Answer: 7 — Explanation: Subtract 8 from both sides to get $5n = 35$, then divide by 5 to get $n = 7$.

GED-MATH1-011 — Correct: A — Correct Answer: 3:4 — Explanation: The ratio 18:24 simplifies by dividing both numbers by 6, giving 3:4.

GED-MATH1-012 — Correct: A,B — Correct Answer: $2x + 5 = 17 \mid 3x - 4 = 14$ — Explanation: Substituting $x = 6$ makes A and B true: $12 + 5 = 17$ and $18 - 4 = 14$.

GED-MATH1-013 — Correct: C — Correct Answer: 45 — Explanation: Add the values: $42 + 38 + 55 + 45 = 180$. Then divide by 4 to get 45.

GED-MATH1-014 — Correct: 28 — Tolerance: exact — Correct Answer: 28 — Explanation: Substitute $x = 12$: $y = 3(12) - 8 = 36 - 8 = 28$.

GED-MATH1-015 — Correct: C — Correct Answer: 42 miles — Explanation: Multiply 3.5 inches by 12 miles per inch to get 42 miles.

GED-MATH1-016 — Correct: B — Correct Answer: $30 + 0.10t = 47$ — Explanation: The total bill is the \$30 monthly fee plus \$0.10 times the number of texts.

GED-MATH1-017 — Correct: A,C — Correct Answer: The volume is 72 cubic feet. | The area of the base is 24 square feet. — Explanation: The base area is $6 \times 4 = 24$ square feet, and the volume is $24 \times 3 = 72$ cubic feet.

GED-MATH1-018 — Correct: B — Correct Answer: $2x \geq 18$ — Explanation: “Twice a number” is $2x$, and “at least” means greater than or equal to.

GED-MATH1-019 — Correct: B — Correct Answer: 72 — Explanation: In order, the values are 68, 70, 72, 75, 80, so the middle value is 72.

GED-MATH1-020 — Correct: 10 — Tolerance: exact — Correct Answer: 10 — Explanation: Add 9 to both sides to get $4x = 40$, then divide by 4 to get $x = 10$.

GED-MATH1-021 — Correct: B — Correct Answer: 35 meters — Explanation: For a rectangle, $2l + 2w = 190$. Substitute $l = 60$: $120 + 2w = 190$, so $2w = 70$ and $w = 35$.

GED-MATH1-022 — Correct: B — Correct Answer: $900(0.85)^2$ — Explanation: A 15% decrease leaves 85% of the value each year, so after 2 years the value is $900(0.85)^2$.

GED-MATH1-023 — Correct: B — Correct Answer: $3/10$ — Explanation: There are 10 marbles total and 3 are blue, so the probability is $3/10$.

GED-MATH1-024 — Correct: B — Correct Answer: $y = 4x + 3$ — Explanation: The y-values increase by 4 each time x increases by 1, and when $x = 1$, $y = 7$, so the equation is $y = 4x + 3$.

GED-MATH1-025 — Correct: 51.84 — Tolerance: exact — Correct Answer: 51.84 — Explanation: The tax is $0.08 \times 48 = 3.84$, so the total is $48 + 3.84 = 51.84$.

GED-MATH1-026 — Correct: C — Correct Answer: 12 — Explanation: Add the equations to get $2x = 24$, so $x = 12$.

GED-MATH1-027 — Correct: A — Correct Answer: 13 feet — Explanation: Use the Pythagorean theorem: $5^2 + 12^2 = 25 + 144 = 169$, and $\sqrt{169} = 13$.

GED-MATH1-028 — Correct: D1=19 — Correct Answer: D1=19 — Explanation: $f(3) = 2(3^2) + 1 = 2(9) + 1 = 19$.

GED-MATH1-029 — Correct: C — Correct Answer: \$16.00 — Explanation: Multiply 2.5 by \$6.40 to get \$16.00.

GED-MATH1-030 — Correct: A,B — Correct Answer: The slope is -2 . | The y-intercept is 9. — Explanation: In $y = mx + b$, m is the slope and b is the y-intercept.

GED-MATH1-031 — Correct: C — Correct Answer: 120 — Explanation: Divide 840 boxes by 7 truckloads to get 120 boxes per truckload.

GED-MATH1-032 — Correct: D — Correct Answer: 21 — Explanation: Distribute: $3y - 12 = 2y + 9$. Subtract $2y$ and add 12 to get $y = 21$.

GED-MATH1-033 — Correct: C — Correct Answer: 502.4 cm^3 — Explanation: Volume = $\pi r^2 h = 3.14(4^2)(10) = 3.14(16)(10) = 502.4 \text{ cm}^3$.

GED-MATH1-034 — Correct: B — Correct Answer: $h = 2A/b$ — Explanation: Multiply both sides by 2 to get $2A = bh$, then divide by b to get $h = 2A/b$.

GED-MATH1-035 — Correct: C — Correct Answer: 6 months — Explanation: Set the costs equal: $18 + 7m = 10m$. Then $18 = 3m$, so $m = 6$.

GED-MATH1-036 — Correct: C — Correct Answer: 10,750 — Explanation: The order has $36 \times 500 = 18,000$ sheets. After using 7,250, $18,000 - 7,250 = 10,750$ sheets remain.

GED-MATH1-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: $T1=y = 5x + 2$; $T2=y = 2x + 5$; $T3=y = -5x + 2$ — Explanation: In $y = mx + b$, m is the slope and b is the y-intercept.

GED-MATH1-038 — Correct: C — Correct Answer: 10% — Explanation: The increase is $748 - 680 = 68$. Then $68 \div 680 = 0.10$, or 10%.

GED-MATH1-039 — Correct: B — Correct Answer: $2x - 5 = -11$ — Explanation: Substituting $x = -3$ gives $2(-3) - 5 = -6 - 5 = -11$.

GED-MATH1-040 — Correct: A — Correct Answer: 28.8 inches — Explanation: After the first bounce the height is $80(0.60) = 48$. After the second bounce it is $48(0.60) = 28.8$.

GED-MATH1-041 — Correct: A,B — Correct Answer: The range is 4. | The median is 4. — Explanation: The range is $6 - 2 = 4$. The middle two values are 4 and 4, so the median is 4.

GED-MATH1-042 — Correct: D — Correct Answer: 23 — Explanation: $g(-2) = -4(-2) + 15 = 8 + 15 = 23$.

GED-MATH1-043 — Correct: B — Correct Answer: The fixed cost before making any notebooks — Explanation: In $C = 120 + 2.75n$, 120 is the fixed cost that does not depend on the number of notebooks.

GED-MATH1-044 — Correct: A — Correct Answer: x : 0, 1, 2, 3
 y : 4, 10, 16, 22 — Explanation: In choice A, each y -value increases by 6 when x increases by 1.

GED-MATH1-045 — Correct: C — Correct Answer: $s = p/0.08$ — Explanation: From $p = 0.08s$, divide both sides by 0.08 to solve for s .

GED-MATH1-046 — Correct: C — Correct Answer: \$2,112 — Explanation: The area is $32 \times 24 = 768$ square feet. The cost is $768 \times \$2.75 = \$2,112$.